P2B | Consolidation of the Real and Digital Twins of the R3 Component

Groups 3R31, 3R32 and 3R33 | PPCA/University of Brasília — IT Project Management

ENGLISH CONTROLLED WORKING DRAFT — NOT THE COMPLETE TRANSLATION | Source: Portuguese V07, 25 October 2026

Status: controlled English technical adaptation aligned to Portuguese V08; equivalence and source evidence subject to final editorial validation.

Executive summary

This document critically integrates three P2A group reports addressing the R3 solid-waste recycling component, including waste-picker cooperatives, and its prospective digital twin. It distinguishes the real sociotechnical system (RSRSR3), which performs collection, sorting, material and organizational transformation, and commercialization, from its digital counterpart (DSRSR3), which represents the real system, contextualizes information, evaluates alternatives and supports human decisions. The proposal combines the four F/S/P/C dimensions (Function, Structure, Process and Context), six pairwise relations, four three-dimensional faces, and a full tetrahedral synthesis for each system. The real and digital components are expected to become progressively coupled through faithful representation, synchronization, controlled intervention and measured feedback.

The proposed initial four-agent architecture is not derived from an assumption that four agent categories necessarily require four executable agents. The project compares three-, four-, and five-or-more-agent alternatives, identifies distinct responsibilities, and selects four as a revisable initial target. All significant real-world decisions remain under authorized human control. The scope is documentary and architectural: neither operational deployment nor empirical superiority of the proposed architecture has been demonstrated.

1. Purpose, project scope and deliverables

The objective is to provide a consolidated and traceable F/S/P/C specification of RSRSR3 and DSRSR3, including real–digital interfaces, agent capabilities, architectural alternatives, risks, feasibility filters and reasons for decisions. The scope covers C1 selective collection, T1 sorting, T2 material and organizational transformation, and C2 commercialization. CIR/DF is used as a bounded documentary example rather than a universally representative field validation.

Official P2B deliverable	Requirement
Portuguese and English posters	Two PDFs, one page each

Portuguese and English technical reports	Two PDFs, 20–30 pages each, using the final-report organization required by the instructor with reference to PMBOK 6 §4.7.3.3
Portuguese and English lessons learned	Two PDFs, one or two pages each

2. Consolidation method and source provenance

The three inputs are P2A reports from groups 3R31, 3R32 and 3R33. The 3R32 report was used as an initial organizational baseline, without epistemic priority: stronger or corrective evidence from either other group must prevail. Official instructor requirements take precedence over group-specific terminology. Consolidation proceeded through three iterative levels: (i) initial description; (ii) enrichment using documented scientific and institutional information; and (iii) selection, adjustment or rejection based on feasibility and higher-level-system constraints. The tetrahedral representation then examines vertices, edges, faces and overall synthesis.

The 3R31 report contributes explicit tetrahedral iteration and sociotechnical/governance considerations; 3R32 contributes architectural decomposition, traceability, decision boundaries and a complementary 2026 bibliometric protocol; 3R33 contributes operational digital stages, real–digital evolution and CIR/DF as a documentary reference. These are source characterizations, not claims of collective endorsement.

2.1 Bibliometric evidence: two complementary, non-additive strands

Group 3R31 reports a Web of Science lexical search concerning recycling and F/S/P/C, including 59,765 records in its broad recycling query and 14 records in the simultaneous four-dimension lexical query. These are retrieval counts, not proof that only fourteen studies model all four dimensions. Group 3R33 reuses and interprets the previously reported 2025 searches and explicitly states that it did not repeat them independently. Group 3R32 reports three separate searches performed on 19 October 2026 with 291, 59 and 8 hits, yielding a within-protocol deduplicated union of 349 UT identifiers. The resulting sets must not be arithmetically pooled without common metadata, documented search comparability and cross-set deduplication. A conclusion about the absence of publications must remain bounded by the materials actually screened.

2.2 AI assistance and human decisions

GPT-6, called “Astra” within the team context, assisted document examination, comparison matrices, synthesis proposals, architectural analysis and writing. The user coordinating consolidation individually approved decisions P2B-01 through P2B-11. Such approval does not constitute endorsement by all twelve participants, by the instructor, or by the cooperatives. Audit trails must distinguish direct source evidence, AI-derived inferences, and human authorizations.

Application of the three iterative levels

This section expands the controlled English draft against the approved Portuguese baseline. The three iterative levels required for P2B are a description-development method rather than an implementation-maturity model. Level 1 characterizes the initial component and its boundaries. Level 2 enriches the description with documented research, technological alternatives and stakeholder knowledge if available. Level 3 screens configurations against higher-level constraints and records which options are adopted, conditional, revised or excluded. Each level may trigger revisions of earlier F/S/P/C relationships.

Dimension	Level 1: initial model	Level 2: enrichment	Level 3: configuration screening
Real F	Materials recovery; environmental, social and economic value	Additional measurable objectives and potential value pathways	Align performance ambitions with cooperative priorities and feasible capacities
Real S	People, facilities, tools, knowledge and organizational links	Possible arrangements, instrumentation and infrastructure	Check cost, availability, occupational safety and self-governance
Real P	C1 collection, T1 sorting, T2 material/organizational transformation and C2 commercialization	Detailed material, information and financial flows; feedback loops	Reject unsupported measurements and infeasible operational changes
Real C	Waste system, P3RD, market and institutional setting	Interdependencies and legal, territorial, economic and social interfaces	Maintain compatibility with upper-level system constraints
Digital F	Represent the real system and support understanding	Scenario analysis, forecasting under assumptions, decision reference	Limit functions to validated data/model availability and authorized use
Digital S	Records, models, services, interfaces	WHAT/HOW capabilities and candidate agents	Check infrastructure, permissions, costs, security and maintainability
Digital P	Acquire, qualify and contextualize information	Simulate, compare, justify, present and learn in a controlled way	Require validation gates, human review, exception handling and rollback

Digital C	Data/technology ecosystem and governance	IT/OT/ET convergence and links to other twins	Respect privacy, cybersecurity, policy and institution-specific constraints

Provenance controls and scope of bibliometric claims

The three source reports do not contribute three independent commensurable bibliometric corpora. The 3R31 report documents a 2025 Web of Science lexical F/S/P/C search, including a broad recycling result of 59,765 and a four-dimension result of 14. These are retrieval counts tied to a particular query and time, not definitive counts of conceptually integrated studies. The 3R33 report reports reuse and interpretation of prior 2025 searches, not a fresh independent query, and distinguishes papers read in full from papers considered on abstract evidence. The 3R32 report describes queries performed on 19 October 2026 yielding 291, 59 and 8 initial results and 349 unique UT identifiers after internal deduplication. Those results address different search questions and cannot be arithmetically pooled with 3R31 counts.

Claims of absence of scholarship must be limited to the actual examined sets. No independent cross-corpus deduplication or new live Web of Science replication is claimed here. The source dossiers should retain query strings, execution dates, filters, export fields, unique IDs and the analysis performed, where available. If the original raw exports are unavailable, the reported counts must be described as reported in the P2A source rather than independently reproduced.

Consolidation decisions and fidelity to the Portuguese source

This section supplements the topic-level English draft with explicit operational assumptions, evidence boundaries and decision procedures carried in the Portuguese V07. It does not claim sentence-by-sentence equivalence; this version remains a controlled technical draft pending bilingual reconciliation.

Scope boundaries and traceability

The unit of analysis is the R3 recycling component as a whole, covering C1 selective collection, T1 sorting, T2 material and organizational transformation, and C2 commercialization. It is not limited to the sorting process or to a single cooperative.

The real system and its digital twin have distinct Function, Structure, Process and Context (F/S/P/C). Their convergence is pursued through interoperable data, human decisions and controlled real-world feedback, not by equating physical assets with software constructs.

The project consolidates three P2A submissions into a documentary, conceptual technical report. It does not demonstrate operational deployment, interviews, field observations, validated forecasts, or actual cost savings unless supported by primary evidence.

For every substantive imported assertion, the final evidence ledger should identify the originating group and document, page, relevant supporting passage, integration action (adopt / adapt / defer / reject), reasoning, and human decision status.

Four agent types in a taxonomy do not automatically imply an optimal population of four executable agent instances. The number remains an initial architecture hypothesis, subject to assessment against alternatives.

Three iterative levels applied to both systems

Level	Real RSRSR3	Digital DSRSR3
Level 1 — baseline	Define human, organizational and physical resources; map C1–T1–T2–C2 and context boundaries.	Define initial conceptual model, data sources, services, and differences from the real system.
Level 2 — enrichment	Use documented literature and stakeholder perspectives where evidence exists; explore alternate workflows and organizational configurations.	Add WHAT/HOW capabilities, controlled agents, simulations, interoperable contracts and candidate human decision journeys.
Level 3 — viability screening	Consider budget, safety, governance, facilities, skills, legal requirements and constraints of higher-level systems.	Condition or remove capabilities unsupported by data, authorization, tested models or feasible infrastructure; record excluded options.

The levels are iterative: newly identified constraints may require reworking a Function, Structure, Process or Context claim and revisiting preceding decisions. They are distinct from the four proposed technical-maturity stages for the digital twin.

3. Real system — RSRSR3

Dimension	Consolidated specification
Function	Recover and recirculate materials while generating environmental, social and economic value and supporting the socioproductive inclusion of waste pickers.
Structure	Cooperatives, people, tacit and formal knowledge, equipment, facilities, materials, organizational rules, governance, contracts and institutional relationships.
Process	C1 selective collection, T1 sorting, T2 material and organizational transformation, and C2 commercialization, with material, information and financial flows and feedback loops.

Context	The wider solid-waste system, the P3RD context, public policy, regulation, markets, communities, social participation and higher-level system constraints.

Tetrahedral interpretation: F/S/P asks whether people and facilities can perform the work required by the intended outcomes; F/S/C examines whether resources and organization can realize the function under institutional conditions; F/P/C considers how process changes affect system purposes and the external environment; S/P/C checks whether physical and organizational resources support feasible processes. Increasing collection alone is not necessarily a system improvement when downstream sorting or commercialization is constrained.

Tetrahedral decomposition: six edges and four faces

The F/S/P/C tetrahedron must not collapse into four independent lists. For both systems, its six edges test pairwise coherence. F–S asks whether available structure can deliver the desired function; F–P examines whether processes actually realize that function; F–C tests the fit between purpose and external constraints. S–P examines resources needed to execute processes; S–C considers organizational and technological compatibility; P–C examines external dependencies and constraints on workflows.

Face	Real recycling system — coherence question	Digital twin — coherence question
F/S/P	Can people, equipment and organizational knowledge carry out C1/T1/T2/C2 to create the intended benefits?	Can information, models, services and agents produce traceable representations and useful analyses?
F/S/C	Are the intended benefits compatible with cooperative self-management, available facilities, markets and regulation?	Can the digital mission be delivered with feasible infrastructure, permitted data and accountable governance?
F/P/C	Do collection, sorting, transformation and sales remain consistent with system-wide and market requirements?	Do ingestion, simulation and recommendation respect privacy, institutional rules and human decision authority?
S/P/C	Are workforce, equipment, contracts and actual operating flows mutually consistent?	Are agent permissions, software services, audit trails, interfaces and failure responses mutually consistent?

The integrated tetrahedral synthesis does not claim that material operations and digital calculations become ontologically identical. Rather, the proposed end state is a coordinated sociotechnical system with bidirectional information and feedback: observed real events populate controlled digital representations; models generate scenario

analyses; competent humans decide; authorized interventions affect the real system; independently recorded outcomes return to the twin. This cycle requires progressive fidelity, synchronization and interoperability, not merely visualization.

Four-face synthesis and cross-system dependencies

Face	Real-system implication	Digital-system implication
F–S–P	Resource, skills and organizational arrangements must enable the four process families to achieve recovery and socio-economic value; sorting throughput alone is insufficient.	Data, models and services must perform verifiable information processes that support explanations and decision-making.
F–S–C	A cooperative's purpose and structure must jointly respect labor safety, self-management, financing and local institutional context.	The digital purpose must remain feasible within infrastructure, access-control, legal and interoperability conditions.
F–P–C	Changes in collection or transformation need to be assessed for downstream consequences and external obligations.	Digital recommendations must explicitly represent institutional constraints and preserve human approval of consequential real actions.
S–P–C	Equipment, workforce, facilities, contracts and materials flows must remain compatible under external constraints.	Agents, APIs and workflows must enforce permissions, audit logging, versioning, failure handling and graceful rollback.

The flattened tetrahedral representation makes all four three-dimensional faces simultaneously reviewable. The synthesis must nevertheless restore all four dimensions as an integrated whole, with feedback across faces. The proposed construction is a method for consistency checking rather than a measured geometrical representation.

Operational real–digital interfaces

Real → Digital: Observed weights, material batches, events, stocks and authorized organizational records enter with source, unit, timestamp, owner and validation status.

Digital → Humans: Dashboards, scenarios and ranked alternatives are accompanied by assumptions, uncertainty, provenance and an explicit opportunity to reject or request revision.

Humans → Real: Legitimate decision-making bodies authorize consequential changes; approval is recorded separately from an agent proposal.

Real → Digital feedback: Observed outcomes after an intervention are checked against predictions and may trigger controlled changes to the digital reference model.

Digital ↔️ adjacent digital twins: Interfaces expose only authorized and contextually appropriate information under agreements and enforceable restrictions.

4. Digital twin — DSRSR3

Dimension	Consolidated specification
Function	Provide faithful and increasingly timely representations of R3 and defensible, traceable references for human decisions regarding possible future configurations.
Structure	Contextualized data, information models, ontologies, interfaces, computing services, simulation components, agent roles A1–A4, access control and governance services.
Process	Acquire and qualify data; contextualize; update digital states; formulate decision questions; explore alternatives; compare scenarios; present recommendations; verify outcomes and revalidate models.
Context	Digital infrastructure, IT/OT/ET convergence, interoperability, security, privacy, regulation, higher-level digital twins, and the institutional capacity of participating organizations.

The eight digital steps are specifications for a proposed solution, not evidence of an operational end-to-end deployment. The DSRSR3 tetrahedron differs from the real system's tetrahedron even when the two form an increasingly integrated sociotechnical system. Its F/S/P face tests whether architecture and computational processes can deliver defensible analyses; F/S/C addresses the fit between decision purpose and institutional/technology constraints; F/P/C limits recommendations to authorized and relevant decisions; S/P/C examines technical interoperability, permissions, versioning and error handling.

Digital operational workflow and interfaces

Eight steps provide a verifiable functional process for the proposed DSRSR3. They are intended capabilities, not assertions of operational deployment.

Step	Proposed digital activity	Required safeguard
1	Acquire records from authorized sources	Document source, permission, units, timestamps and receipt status
2	Qualify and contextualize records by C1, T1, T2 or C2	Detect missing, duplicated, incompatible or uncertain

		entries
3	Update the represented system state	Preserve version, observation-versus-estimate distinction and uncertainty
4	Formulate the decision problem	Specify requester, objective, constraints and permitted decision scope
5	Generate alternatives or simulated outcomes	Document model assumptions, data limitations and feasible domain
6	Compare alternatives using explicit criteria	Check capacity, costs, material recovery, worker safety and social effects
7	Explain recommendations and request human review	Maintain contestability and prohibit implicit approval
8	Record verified outcomes and update models under control	Version changes, document validation and rollback options

Four real-process interfaces illustrate how fidelity might be evaluated. For C1, event records may contain material type, origin category, receipt time and measured amount. For T1, the interface may compare input, sorting outputs, reject streams and stock changes. For T2, it must distinguish physical/material transformation from organizational transformation; these are not interchangeable variables. For C2, commercial transactions, weighed quantities, unit values, timing and contractual context must be reconciled. Each interface remains conditional on legitimate data access and adequate measurement practices.

5. WHAT/HOW capabilities and the agent architecture

WHAT capability	Target responsibility
DS — Data Services	Source quality, contextualization, provenance and reproducibility
IR — Integration	Controlled interfaces among processes, records, equipment and related systems
IC — Intelligence	Indicators, conditional predictions, simulations and scenario comparison
UX — User Experience	Accessible explanations, contestability and role-sensitive interaction
MG — Management / instructor's "Marketing"	Terminological discrepancy requiring explicit editorial resolution rather than silent substitution
TW — Trustworthiness	Security, privacy, reliability, resilience, governance and accountable decisions
HOW capability	Target responsibility
PK — Perception & Knowledge	Acquire and interpret authorized information

CG — Cognition & Reasoning		Frame problems and compare alternatives
LA — Learning & Adaptation		Controlled updating based on validated observations
AE — Action & Execution		Execute permitted digital workflows
IC — Interaction & Collaboration		Coordinate information and agents
GS — Governance & Safety		Constraints, monitoring, security, privacy and human control
Agent role	Distinct responsibility	Autonomy boundary
A1 — Conversational	Explain records and grounded indicators to authorized users	No fabricated figures or unauthorized record changes
A2 — Procedural	Check records and perform explicitly authorized digital routines	No unapproved high-impact action
A3 — Cognitive	Compare scenarios, quantify assumptions and identify infeasibility	No unilateral changes in physical operations
A4 — Collaborative / MAGS	Coordinate cross-process analyses and resolve analytical conflicts	No power to impose collective organizational decisions

5.1 Why initially four agents, rather than three, five or any other number?

The P2A evaluators explicitly asked why four agents had been selected. The answer is not that the four categories A1–A4 force the implementation of exactly four executable agents. Six functional requirements were identified, of which four motivate potentially separate agent roles (communication, procedure execution, cognitive analysis and collaboration); governance and controlled learning are cross-cutting. A three-agent option combines at least two responsibilities; the four-agent option permits a direct initial separation; five or more agents increase specialization while adding interfaces and coordination demands. The four-agent option is thus a reasoned and revisable architecture target, not a proven optimum. Functional, integration and architectural tests will be required to establish whether a different number performs better.

Functional allocation and alternative agent counts

In the P2A presentations, faculty explicitly asked why four agents had been chosen rather than three, five or another number. The decision recorded in P2B-06 is therefore an architectural hypothesis supported by a functional decomposition, not a numerical optimum. A1–A4 name four agent categories in the teaching assignment; categories do not automatically determine the number of deployed software instances.

Candidate	Plausible advantage	Unresolved trade-off
Three agents	Lower nominal component count; conversational and procedural services can be combined	Role combination may concentrate privileges and obscure accountability; collaboration may be merely centralized orchestration

Four agents — preferred initial reference	One clearly accountable lead role for consultation, authorized workflows, scenario reasoning and inter-agent coordination	Additional interfaces must justify their overhead; optimality has not been tested
Five or more agents	Additional specialization by C1/T1/T2/C2, sector, data protection or analytical function	More coordination, privileges and failure modes; extra specialization requires observed need
Agent role	Main input and output	Access and acceptance boundary
A1 conversational	User questions and permitted records → cited answers, indicators and explanations	Read-only within the requester's permissions; test grounded answers and privacy
A2 procedural workflow	Events, records and approved rules → checked tasks, exceptions and process logs	Only explicitly authorized reversible digital workflows; test duplication, exceptions and rollback
A3 cognitive analytical	Validated data, objectives, constraints and models → scenarios, estimates and recommendations	No unilateral physical actions; test feasibility, sensitivity and model uncertainty
A4 collaborative intelligence / MAGS	Agent outputs, inter-process constraints and permissions → coordinated proposals and conflict reports	No authority to impose cooperative agreements; test isolation, conflict resolution and institutional consent

Governance and safety (GS) is a cross-cutting HOW capability, not an automatically necessary fifth agent. Learning and adaptation (LA) is likewise conditional on review and version control, rather than a mandate for autonomous updating. A query about an existing indicator may require A1 alone; a decision involving a sorting bottleneck might employ A2, A3 and A4 before A1 presents the result. The number of invoked agents should therefore depend on the case, not a fixed requirement to activate all four.

A defensible comparison protocol must test the same cases under candidate three-, four- and five-agent designs. It should consider functional coverage, traceable accountability, latency, inter-agent overhead, unsafe requests, privileges, correctness and maintainability. No measured comparative results are available in the P2A materials; the preferred initial four-role design must therefore remain revisable.

Decision cases for testing the proposed agent architecture

Case 1 — sorting bottleneck

A2 checks the consistency and completeness of inbound/outbound lot records; A3 explores feasible capacity scenarios; A4 evaluates upstream collection and downstream commercialization impacts; A1 communicates alternatives and uncertainties to authorized decision makers. A cooperative manager or competent collective body

approves any operational change. Test evidence: traceable inputs, constraints, rejected alternatives, permission checks and decision logs.

Case 2 — mass-balance discrepancy

The system compares incoming material, outgoing fractions, stock variation and documented adjustments over the same accounting period. If the balance is inconsistent, A2 opens a verification task instead of imputing an unobserved weight; A3 may examine possible causes as hypotheses. Acceptance requires accounting consistency within a predeclared, context-specific tolerance and human examination of exceptions.

Case 3 — commercialization options

A3 evaluates documented price, logistics, timing and contractual constraints; A4 identifies cross-cooperative or cross-process dependencies only when permitted; A1 explains trade-offs. Forecasts are not observed transactions. Validation requires evidence for input prices and costs, constraints, alternatives, risk and the human decision.

Case 4 — inter-cooperative data exchange

A4 may coordinate exchange of approved aggregate indicators through explicitly contracted interfaces. Access to personal or commercially sensitive records is not inferred from participation in a multiagent network. Failure tests should include unauthorized requests, inconsistent records, partner disconnection, conflicting objectives and safe fallback.

Comparison with three and five agents

The four-role proposal separates human interaction, rule-governed digital workflows, analytical reasoning and collaborative coordination. A three-agent alternative merges at least two responsibilities; a five-agent alternative introduces specialization needing independent justification. Compare coverage, traceability, workload, interface count, security boundaries, failure propagation and maintainability using common test cases. No configuration is presented as empirically optimal.

Agent-count validation criteria and decision boundaries

The P2A reviewers asked why the design selected four agents rather than three, five, or another number. The P2B consolidation treats this as an architectural research question, not a rhetorical challenge to be dismissed.

Criterion	Three-agent design	Four-agent design	Five-or-more design
Functional coverage	Feasible if at least two responsibilities are combined	Feasible with one agent role per identified responsibility	Feasible with further role partitioning
Separation	Some duties necessarily shared	A1 interface; A2 workflow; A3 analysis; A4 coordination	Potentially more granular but needs justification
Coordination	Potentially lower	Intermediate in	Potentially greater

burden		principle	
Access isolation	Depends on enforcement	Easier to associate core permissions with roles	May improve isolation but multiplies interfaces
Evidence of superiority	Not demonstrated	Not demonstrated; preferred initial hypothesis	Not demonstrated

RF-01: explain authorized operational data; RF-02: check and route procedural work; RF-03: compare constrained scenarios; RF-04: coordinate cross-process options. Governance/security (RF-05) and validated learning (RF-06) are cross-cutting and do not necessarily require dedicated executable agents.

The project preference for four initial roles is thus grounded in responsibility boundaries. Later acceptance requires functional test cases, integration tests and comparative architecture tests. No objective score, optimality claim or tested superiority is asserted in the documentary phase.

A1–A4 authority and failure boundaries

A1 Conversational: Authorized read-only inquiry, grounded answers and uncertainty disclosure. Main hazard: fabricated or confidential answers. Validate against source records, permissions and ambiguous queries.

A2 Procedural workflow: Execute permitted digital processes and report exceptions, without inventing missing material measurements. Main hazard: unauthorized or repeated transactions. Validate preconditions, idempotency and rollback.

A3 Cognitive autonomous: Evaluate constrained alternatives, sensitivity and uncertainty; no unilateral physical intervention. Main hazard: infeasible or overconfident recommendations. Validate model assumptions and scenario reproducibility.

A4 Collaborative intelligence: Coordinate analyses and identify conflicting recommendations while respecting inter-organizational permissions. Main hazard: unauthorized sharing or inconsistent coordination. Validate isolation, conflict resolution and audit trails.

6. Real–digital convergence and verification

Convergence is evaluated in three complementary dimensions: fidelity (agreement with independent real measurements), synchronization (age and latency of data), and integration (whether observation, analysis, authorized decision, intervention, and results verification form a controlled feedback cycle). The project proposes four distinct technological stages: conceptual representation, connected replica, intelligent replica, and integrated sociotechnical system. These are project-defined stages, not an official DTC maturity scale, and they do not replace the instructor's three iterative methodological levels.

Potential indicators include completeness of mandatory data fields, source traceability, cross-process material balance consistency, absolute error against real measurements, time-to-update, proportion of grounded A1 answers, A2 workflow compliance, A3 scenario feasibility, A4 coordination consistency, and human review of recommendations. No universal target threshold is asserted without measurement context. Improvements observed in the recycling operation must not automatically be attributed causally to the digital twin.

Measurement framework and progressive convergence

Convergence is assessed along three non-substitutable dimensions: fidelity (agreement with verified real-world reference measurements), synchronization (how promptly relevant state changes are reflected digitally), and integration (whether authorized observations and decisions circulate reliably across the sociotechnical system). Averages must not hide severe failures in privacy, data quality or authorization.

Evaluation domain	Illustrative measure	Limitation
Representation fidelity	Absolute error between digital and independently measured real quantities; consistency of material balances	Requires common units, appropriate sampling intervals and measurement uncertainty
Temporal synchronization	Delay between a real event and its validated appearance digitally	Acceptable latency depends on decision consequence and source availability
Process coverage	C1/T1/T2/C2 functions represented with documented boundaries	Coverage is not equivalent to correctness or field validation
Data provenance	Share of evaluated records with traceable source, date and responsible origin	A fully documented incorrect record remains incorrect
Decision traceability	Share of recommendations with alternatives, constraints, model version and human disposition	Good records alone do not prove better decisions
Safety and privacy	Access-control tests, policy violations and auditable incidents	No claim of certified compliance without independent evidence
Proposed stage	Meaning	Evidence needed before claiming attainment
1 — Conceptual representation	F/S/P/C model, boundaries, data requirements and preliminary control logic	Reviewed architecture and traceable requirements
2 — Connected replica	Authorized observations systematically update selected represented variables	Measured integrity, timing, reconciliation and access controls
3 — Intelligent replica	Qualified models compare plausible futures and issue	Documented scenario tests, model validation and

	contestable recommendations	specialist evaluation
4 — Integrated sociotechnical system	Approved decisions influence real work and real outcomes feed back into updated digital models	Recorded authorizations, real-world outcomes, monitored incidents and controlled model updates

These four stages are a project-specific proposal, not an official Digital Twin Consortium maturity scale. They must be assessed per function and process, since a cooperative might have connected weighing data yet only a conceptual representation of its commercialization process. Furthermore, improved recovery or income cannot be causally attributed to the digital twin merely because the indicators improve after implementation.

Cross-system data interfaces: scope, evidence and controls

Collection (C1)

Potential collection events include source, location at an appropriate level of aggregation, material category, collection date and measured quantities. The digital representation should retain units, event time, entry time, provenance and the recording party. Direct personal identifiers are not default requirements; where personal data would be necessary, an explicit purpose, lawful basis, access policy and retention rules must be established.

Sorting (T1)

Inbound lot records, material fractions, rejects, stocks and sorting capacity may support consistency checks and bottleneck analysis. The digital process must distinguish observed measurement from calculated estimates, and cannot fill missing measurements with fabricated values. Discrepancies trigger verification by the responsible human operator.

Transformation (T2)

T2 includes material processing and organizational transformation. Consequently, the digital counterpart must represent both measurable physical changes, when documented, and organizational constraints such as skills, roles, working conditions and governance arrangements. Social indicators must not be conflated with measured tonnage.

Commercialization (C2)

Proposed records include material type, quantity, transaction date, pricing conditions, delivery status and contract constraints. Sensitive commercial information should be disclosed only to authorized participants. Simulated revenue must be labeled as a scenario, not as actual cooperative income.

Minimal interoperability contract

Each exchanged record should define a stable identifier, source system, event and recording timestamps, process identifier, measurement unit, quality status, access scope and version. Interface owners, error handling, correction history and synchronization periodicity must be stated before any operational integration is claimed.

Acceptance protocol and limits of claimed maturity

Conceptual-model acceptance

Identify F/S/P/C of real and digital components separately, cover C1/T1/T2/C2, represent the four vertices, six pairwise relations and four triadic faces, and record the integrated synthesis. Demonstrate three iterative description levels and the reasons configurations were retained, constrained or excluded.

Data readiness

Before connecting real records, define data stewardship, source authorization, field semantics, units, completeness, quality checks, correction rules and retention. Data-driven conclusions are conditional on the availability and quality of these inputs.

Governance and safety

Apply least privilege, segregation of duties where warranted, audit logging, incident handling, human approval for high-impact acts, and reversible changes wherever feasible. No proposed agent can autonomously impose physical, contractual or collective decisions.

Progressive maturity gates

Stage 1 documents the conceptual representation; Stage 2 requires verified data connections; Stage 3 adds models and decision support with evaluated limitations; Stage 4 closes an authorized decision–intervention–observation loop. Stages describe a project-specific proposed evolution, not an official Digital Twin Consortium maturity standard; individual processes may progress at different rates.

Metrics and evaluation

Separately assess representation fidelity, event-to-digital latency, traceability, functional correctness, constraint adherence, safety and the observed real-system outcomes. Define acceptance thresholds for each use case before trials; do not infer that improvements in recycling metrics were caused by the digital twin without suitable comparative evidence.

Evidence qualifications

Source report evidence is not equivalent to an independently reproduced experiment; an approved architectural decision is not an implemented system; a scenario is not a real operational observation; and a bibliometric keyword count is not proof of an absence of scientific work.

Validation gates and evidence classification

Conceptual baseline → connected replica: Requires identified and authorized data sources, minimum schema, owners, integration specification, security/privacy risk assessment and demonstrable source provenance.

Connected replica → intelligent replica: Requires measured data quality and update latency, validated model constraints, reproducible tests and transparent treatment of missing information.

Intelligent replica → integrated sociotechnical operation: Requires review of recommendations by competent humans, recorded approval pathways, monitored consequences, appeal/rejection mechanisms and reversibility for permitted actions.

Continuous operation: Requires monitoring for drift, access reviews, incident handling, traceability of model/configuration changes and periodic reconsideration of the human governance arrangement.

Measures of digital quality must not be conflated with recovered tonnage, income or social outcomes in the real system. Causal attribution requires a defensible evaluation design; an observed outcome improvement cannot automatically be credited to the digital twin.

Evidence and assertion categories

- Documented in P2A: explicitly reported in the supplied team document, not independently verified in the field.
- Verified against a primary passage: checked in the actual provided source for textual support.
- Proposed in P2B: conceptual model, decision procedure, architecture choice or evaluation plan synthesized for consolidation.
- Not empirically validated: depends on field evidence, external system access, experiments, institutional acceptance or production use.
- Abstract-screened research: weaker basis for deep claims than full-text assessment; no extrapolation from screened items to all published scholarship.

Interface data objects: C1 — Collection / receipt

Event / object	Conceptual fields	Interpretation / governance constraint
Planned collection	route, date, origin, planned capacity	Planning does not prove material was received
Actual receipt	date, unit, material class, measured mass	Preserve instrument and measurement quality
Discrepancy	planned and actual amounts; available explanation	Issue for human reconciliation

Interface data objects: T1 — Sorting

Event / object	Conceptual fields	Interpretation / governance constraint
Inbound batch	identifier, mass, estimated composition, origin	Only compare outputs within the same measurement boundary
Output by class	class, mass, date, responsible party	Supports later reconciliation and commercialisation
Rejects and estimated losses	category, amount, reason	Avoid fictitious positive balance by omitted exits

Interface data objects: T2 — Material and organisational transformation

Event / object	Conceptual fields	Interpretation / governance constraint
Process change	procedure version, date, responsible party, training	Institutional record is not proof of a causal outcome
Material operation	input, product, loss, energy	Material and energy balances

	consumption if available	where applicable
Organisational change	approved rule, minutes/formal evidence, effective date	Requires institutional authority and consent

Interface data objects: C2 — Commercialisation

Event / object	Conceptual fields	Interpretation / governance constraint
Available batch	type, measured mass, quality, stock	Availability is not a completed sale
Commercial transaction	date, volume, price, documented conditions	Observe commercial confidentiality and role-based access
Settlement / receipt	amount, status, reconciliation	Distinguish accrual and cash when relevant

7. Feasibility decisions and evidence limits

Candidate	Consolidation treatment	Reason
Model the complete C1/T1/T2/C2 chain	Adopt	Required R3 boundary
Include cooperative knowledge and self-governance	Adopt	Core sociotechnical structure
Four agent roles as initial target	Adopt conditionally	Functional decomposition; optimum not demonstrated
Full immediate deployment	Condition	Cost, data, access, skills and infrastructure constraints
Uniform real-time data synchronization	Do not adopt at this stage	Update needs differ by decision and data type
Unsupervised high-impact physical changes	Do not adopt	Human legitimacy, safety and authorization
Use individualized personal data	Strictly condition	Demonstrated necessity, legal basis, minimization and security
Claim current full real–digital integration	Do not adopt	No documented operational validation

Material open gaps are quantifiable deployment and operating costs, formal institutional agreements for data access and participation, and measured comparison between real observations and digital representations. All feasibility classifications remain conceptual/architectural until tested against real organizational constraints.

Viability screening and explicit choices

Alternative	Proposed disposition	Rationale / missing evidence
Model C1, T1, T2 and C2 jointly	Adopt for conceptual scope	Avoid an artificially narrow sorting-only twin
Represent cooperatives' human and organizational knowledge	Adopt	The real component is sociotechnical, not only equipment and materials
Four agents as initial	Adopt provisionally	Clear role boundaries; still

architecture		requires comparative evidence
Deploy every integration and every agent immediately	Conditional	Infrastructure, data access, cost and competence not established
Require real-time synchronization for every variable	Do not adopt as a universal rule	Decision cadence and value differ across variables
Allow agents to approve material real-world changes	Do not adopt at this stage	Human accountability and legitimate authority must be preserved
Simulate flow and capacity scenarios	Conditional	Requires validated quantities, assumptions and model behavior
Use individualized worker records	Strictly conditional	Necessity, lawful basis, minimization and safeguards must be established
Enable cross-cooperative agent collaboration	Conditional	Requires sharing agreements, permissions and institutional legitimacy
Assert deployment of a fully integrated twin	Do not assert	No operational integration evidence is provided in the examined P2As

The third iterative level therefore changes the candidate design rather than simply listing constraints. For instance, universal real-time synchronization is replaced by data-specific intervals; unrestricted digital autonomy is replaced by controlled analytical assistance; and simultaneous full deployment is replaced by staged acceptance gates. Quantitative feasibility remains unproven because comprehensive cost estimates, cooperative agreements and independently verified operational measurements have not been supplied.

Consolidated risk register — correspondence with the Portuguese report

Risk	Possible impact	Proposed treatment
Missing or inconsistent operational data	Misleading digital states and recommendations	Source trails, validation, explicit unknown states and safe blocking
Exposure of personal data	Rights violations and potential harm	Data minimisation, segregation, role-based permissions and audit
Operationally infeasible recommendation	Financial loss, waste, and risks to cooperative members	Verified constraints, sensitivity analysis and human veto
Outdated digital representation	Action based on obsolete information	Timestamps, validity periods, refresh before consequential decisions
Uncontrolled agent autonomy	Unauthorized physical or contractual actions	Least privilege, explicit authorisation and rollback provisions
Incompatible interfaces	Broken data lineage and incomplete process coverage	Interface specifications, semantic mapping and exception handling
Assuming documentary evidence is field validation	Overstated maturity and unsupported conclusions	Separate conceptual specifications, documentary checks and empirical validation

Failure-mode controls — acceptance and reversal

Failure mode	Potential consequence	Project response
Missing/inconsistent weighing data	Digital state diverges from the real system	Flag data quality; block definitive inferences
Excessive synchronisation latency	Recommendation based on an outdated state	Display data age; refresh before approval
Unvalidated analytic model	Unsafe or infeasible recommendation	Independent review; baseline scenario
Unauthorised personal-data disclosure	Privacy or rights violation	Access segmentation, minimisation and incident response
Duplicate procedural execution	Incorrect inventory or repeated business transaction	Idempotency checks, logs and reversal workflow
Conflicting agents or cooperative objectives	Misleading consensus or unintended organisational effects	Escalation, explicit disagreement and human resolution
Misinterpreted mass-balance discrepancy	False attribution of losses or gains	Verify measurement boundaries, stock variation and instruments

8. Final-report performance assessment — documentary phase

Scope: the documentary P2B consolidation, not a deployment of the digital twin. Quality: source-supported descriptions, explicit methods and documented limitations. Schedule: submission deadline supplied as 30 October 2026 at noon, Brasília local time; interim internal draft progress must not be treated as a complete submitted deliverable. Cost: no reliable project cost baseline or actuals have been provided. Benefits: methodological consistency, transparent integration of three group reports and improved justification of the initial agent count; no measured operational benefit is claimed. Outstanding acceptance conditions: complete bilingual artifacts, source and bibliographic audit, editorial conformance, final participant review, and submission.

Controlled phase-performance assessment: scope covers the documentary consolidation and six required PDF deliverables; quality covers factual traceability, bilingual consistency and independent validation limits; schedule covers the 30 October 2026 noon Brasília deadline without assuming completion; costs remain undocumented without a baseline or actual expenditure records; benefits are methodological rather than demonstrated improvements in cooperative productivity; risks include remaining source checks, PT–EN equivalence, missing field validation and collective review. These categories refer to distinct assessment dimensions and must not be conflated.

Closing-report assessment under PMBOK 6, Section 4.7.3.3

The academic assignment asks for a final-report format consistent with the PMBOK 6 Close Project or Phase outputs. This assessment concerns the documentary consolidation phase, not commissioning of a physical/digital production system.

Assessment category	Current documentary position	Evidence limit
Scope	Integrated real/digital F/S/P/C, interfaces, agents, scenarios and evaluation	No claim that the physical/digital solution was deployed

	framework specified	
Quality	Editorial source checks, conceptual review and explicit open issues	Full literature-citation and empirical validation remain incomplete
Schedule	Submission target stated in the course assignment for 30 October 2026 at noon Brasília time	Detailed baseline-to-actual schedule data were not provided
Cost and resources	Work product uses the three groups' documents and AI-assisted drafting	No validated deployment cost, financial baseline or budget variance
Benefits	Traceability and explicit architectural trade-offs improve the documentary proposal	Operational, social and financial benefits are prospective and not measured
Risk and open issues	Data permissions, economic feasibility, institutional validation, model accuracy, integration and number of agents	Risk mitigation design is not evidence that all risks are controlled
Lessons learned	Agent taxonomy versus instance count; non-additive bibliometrics; preservation of human authority	These are learning outcomes of the academic consolidation process

9. Lessons learned and decision provenance

A classification of agent capabilities cannot by itself justify a number of executable agents; a design must compare architectural alternatives. A document-based case cannot be generalized to all cooperatives without appropriate evidence. Bibliometric counts based on different searches must not be added. Human and organizational knowledge is a structural component of the real system. A digital replica gains decision value when fidelity and governance coexist with safe analytical autonomy; a high-fidelity representation alone is not proof of improved real-world outcomes. Future presentations and posters must make the question "Why four agents?" and the sequence of consolidation decisions explicit.

Decision provenance and attribution

The user individually approved conceptual and methodological decisions P2B-01 through P2B-11 during the consolidation. This is neither collective approval by all twelve group members nor stakeholder field endorsement. GPT-6 (named "Astra" in team discussion) assisted with comparing reports, proposing changes, organizing the decision records and producing text. Named human participants remain responsible for the academic submission and any subsequent adoption decisions.

Decision cluster	What changed	Why the change matters
P2B-04	Unified real F/S/P/C, preserving organizational knowledge and the whole C1–T1–T2–C2 chain	A synthesis, rather than three disconnected summaries
P2B-05	Digital replica fidelity and strategic reference made complementary; progressive supervised autonomy	Real–digital convergence without category confusion
P2B-06 / 07	Four initial agents justified by distinct roles, permissions and testable boundaries	Addresses the faculty question: why four rather than three, five or X?
P2B-08	Convergence measured by fidelity, synchronization and integration with evidence gates	Replaces untestable progress claims
P2B-09 / 10	Three iterative description levels, tetrahedral faces and explicit feasibility dispositions	Turns descriptive lists into interdependent engineering decisions
P2B-11	Separate bibliometric strands, restrict unsupported absence claims and identify evidence maturity	Prevents false precision and overclaiming

Decision rationale and presentation-ready provenance

P2B-04 — Consolidated real-system F/S/P/C

The initially separate group descriptions were reconciled into a socio-technical model spanning the complete R3 processes. The 3R32 report supplied a working backbone; the 3R33 report strengthened distinctions between ends and means and mass-balance considerations; the 3R31 report reinforced organizational knowledge, rights and governance. The approval was individual and should not be misreported as a vote of twelve members.

P2B-05 — Convergence with distinct F/S/P/C

Discussions with the instructor, as reported by the user, established a trajectory toward an increasingly faithful digital replica integrated with real operations. Decision-reference capabilities are complementary rather than a replacement for faithful representation. Operational integration retains separate material and digital roles.

P2B-06/07 — Why four agents?

In the P2A presentations, instructors challenged why exactly four agents were selected. The revised rationale explicitly distinguishes four *agent types* in the assignment from four executable *instances*. Four candidate functional responsibilities motivated a

provisional four-agent architecture; alternatives with three and five or more remain valid until evaluated. Agent permissions, selective invocation and human authorization are specified as controls.

P2B-08/09/10 — Validation, iteration and feasibility

Fidelity, synchronization and integration were adopted as distinct criteria; four progressive technical stages were kept distinct from the instructor's three methodological iteration levels. The tetrahedral faces and selection of feasible alternatives make architecture choices contestable and traceable.

P2B-11 — Bibliometric integrity

The historical F/S/P/C-oriented searches described by 3R31 and reused by 3R33 must not be added to the separate 2026 3R32 search. Reused queries do not constitute a third independent dataset. Literature-gap statements are limited to examined evidence, and source-level verification gaps are disclosed.

Attribution

GPT-6 (called Astra by the project team) assisted with cross-document synthesis, comparative recommendations, drafting and audit structure. Material choices P2B-01 to P2B-11 were approved by the user; this does not establish independent endorsement by the other eleven participants or field stakeholders.

Individual human-approved consolidation decisions — P2B-01 to P2B-11

Decision ID	Approved direction	Reason for choosing it
P2B-01	Critical consolidation procedure, provenance and transparent AI assistance	Prevent uncritical merging and false collective authorship
P2B-02	Operational review order 3R32 → 3R33 → 3R31	Efficient starting point without overriding stronger evidence from another group
P2B-03	Consolidate the real RSRSR3 F/S/P/C first	Establish real-world boundaries before digital specification
P2B-04	Approve integrated real-system F/S/P/C	Preserve people, autogovernance, material flows and context
P2B-05	Progressive real–digital fidelity, integration and supervised autonomy	Combine replica and decision reference while keeping distinct F/S/P/C
P2B-06	Four initial agent roles, subject to comparison with 3 and 5+	Justify quantity by responsibilities rather than taxonomy alone
P2B-07	Functional inputs, outputs, permissions, interactions and tests for A1–A4	Make role boundaries and future validation observable
P2B-08	Convergence metrics, four proposed maturity stages and validation gates	Provide measurable criteria without fabricating implementation
P2B-09	Three iterative levels and tetrahedral relationships	Show systematic enrichment and feasibility screening
P2B-10	Four tetrahedral faces and preliminary feasibility screening	Explicitly link structural and institutional constraints to designs
P2B-11	Dual bibliometric strands,	Avoid duplicate counts and claims

	evidence qualifications and final tetrahedral synthesis	beyond reviewed records

Audit qualification: this register records decisions approved by the individual consolidation lead. It does not imply collective approval by all twelve members, empirical validation, or deployment of the digital twin.

10. Source register and limitations

[G31] Group 3R31, original Portuguese P2A technical report, 18 pages. [G32] Group 3R32, original Portuguese P2A technical report, 28 pages. [G33] Group 3R33, original Portuguese P2A technical report, 29 pages. These original reports are the basis for the statements attributed to groups. Full per-claim page-level verification and a final bibliographic audit remain necessary before submission.

Editorial audit note (EN V10): this English report remains a controlled adaptation of PT V08, not a verified paragraph-by-paragraph translation. Source-backed claims, table alignment and editorial coverage are evaluated in the internal crosswalk. No operational deployment or field validation is claimed.

Controlled bilingual terminology and translation audit

Portuguese baseline	Controlled English rendering	Interpretation rule
Função / Estrutura / Processo / Contexto	Function / Structure / Process / Context (F/S/P/C)	Maintain dimension identity between systems while preserving different contents
Sistema real RSRSR3	Real R3 recycling system (RSRSR3)	Human and physical operations, not the digital representation
Gêmeo digital DSRSR3	Digital twin of R3 (DSRSR3)	Proposed data/model/agent system; do not imply deployed status
Réplica → referência	Replica → Reference	Growing fidelity supports, rather than excludes, future-oriented decision reference
Patamares iterativos	Iterative description levels	Three methodological levels; distinct from four technology stages
Arestas / faces / planificação	Edges / faces / tetrahedral net (unfolding)	No omission of relationships among vertices and faces
Autonomia analítica supervisionada	Supervised analytical autonomy	Human authorization over consequential changes
Rastreabilidade	Traceability / provenance (as appropriate)	Traceability of decisions versus provenance of source records
Viabilidade condicionada	Conditional feasibility	Not a verified economic or

		operational result
Lições aprendidas	Lessons learned	An academic/project closeout output, not a substitute for field evaluation

Terminology caveat: teaching material mentions MG as “Marketing”, whereas source technical discussions also use “Management”. This discrepancy must be cited and resolved explicitly in the final WHAT mapping rather than silently translating both as the same technical family. Both labels must be preserved in the audit record until primary-source alignment is complete.

Editorial audit note (EN V10): this English report remains a controlled adaptation of PT V08, not a verified paragraph-by-paragraph translation. Source-backed claims, table alignment and editorial coverage are evaluated in the internal crosswalk. No operational deployment or field validation is claimed.

Provenance, bibliometrics and report-performance boundaries

The 3R31 bibliometric axis searches lexical intersections of Function/Structure/Process/Context. The 3R33 documentary work revisits historic searches rather than independently replicating the same querying operation. The 3R32 October 2026 search applies a distinct digital-twin and cooperative-decision framing and reports within-query deduplication. Counts from those strands must not be summed as if they arose from a single corpus.

The final-report assessment covers documented project deliverables, quality checks, schedule information, known or unknown project expenditure, expected benefits, open risks and limitations. Unavailable actual expenditures or intervention benefits must be explicitly classified as not documented, rather than replaced by assumed amounts.

The report is prepared with assistance from GPT-6 (called “Astra” by the project team). Source interpretation, drafting and comparison can be attributed to the AI; approval of P2B-01 through P2B-11 belongs to the human consolidator and cannot be described as a vote or ratification by all twelve group members.

Editorial audit note (EN V10): this English report remains a controlled adaptation of PT V08, not a verified paragraph-by-paragraph translation. Source-backed claims, table alignment and editorial coverage are evaluated in the internal crosswalk. No operational deployment or field validation is claimed.

Appendix — bilingual evidence-control and table concordance

This appendix records unresolved equivalence limitations between Portuguese baseline V08 and English V10. It is an editorial quality-control record, not an additional experimental result.

Portuguese content	English status	Required editorial treatment

Six relationships of the real-system tetrahedron	Six relationships are described in prose; dedicated aligned table remains to be checked	Verify every pair for both real and digital twins
Risk and failure-mode registers	Aligned risk and failure-mode tables inserted into EN V08	Compare individual safeguards and wording against PT source
Interface data objects for C1, T1, T2, C2	Dedicated C1, T1, T2 and C2 interface tables inserted	Perform final field-by-field and interpretation review
P2B-01 through P2B-11 decision register	Eleven individual decision rows inserted into EN V08	Verify approval wording and rationale against PT and conversation
Glossary and evidentiary qualifiers	Partial English terminology table	Maintain documented / proposed / primary-source-checked / not empirically validated distinctions.
Bibliometric claims for 2025 and 2026	Described as complementary studies	Do not sum samples or claim independent 3R33 search.
Why four agents, rather than three or five?	Dedicated section retained	Keep as an architectural hypothesis; no optimality claim.

Reporting rule: a checked page anchor is not, by itself, proof of semantic support; and a described capability is not evidence of implementation. Claims about stakeholders, costs and cooperative field evaluation remain conditional unless independently evidenced.

Editorial audit limitation (internal): thematic correspondence between the Portuguese and English reports does not demonstrate sentence-by-sentence translation equivalence. Source-page validation and independent bibliographic verification remain separate acceptance gates.